# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

**MG1DSCCSC100- Art of Computing and Problem Solving**

Duration: 1.5 hrs Maximum Marks: 50

*Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.*

**Part A**

Very Short Answer Questions
Answer All Questions
Each question carries 1 mark

1. If a computer has 16 MB of RAM, how many bytes of data can it hold? [U][CO1]
2. You are using a computer to create a presentation for a meeting while also checking your email. What type of software are you using in each case, and why? [U][CO1]
3. Name the translator that translates source code line by line during execution. [K][CO1]
4. Identify the errors in the following code. [U][CO1]

```
int main();
{
 for (int i = 0, i < 10, i++);
 printf("%f\n", i);
 return 0;
}
```

5. The identifier ‘sum’ is not equal to ‘SuM’. Justify your answer. [U][CO2]
6. Identify a data type in C that signifies the absence of a value or null. [K][CO2]
7. Which header file includes the declaration for the printf() function in C? [K][CO2]
8. Write the output of the following program segment? [A][CO3]

```
int i = 0;
while (i <= 10)
{
printf("%d\n", i+2);
}
```

9. A function is defined to calculate the factorial of a number and return the result to the calling function. What return type should be used in the function prototype? [A][CO3]
10. A program requires checking if two user-entered email addresses are the same during registration. Which string function would you use to compare the two email addresses in C? [A][CO3]

[1 x 10 = 10]

## Part B

Short Answer Questions
Answer **4** Questions
Each question carries **5** marks

11. Discuss the features of high-level languages that make them more user-friendly compared to assembly language. [U][CO1]
12. In a distance conversion program, you are asked to convert a distance from kilometers to miles. The formula for conversion is *miles=kilometers×0.621371*. Using this formula, identify the constants and variables. Explain the role of each in the conversion process. [U][CO2]
13. Draw a flowchart to find the largest among three numbers. [U][CO1]
14. Compare the pre-increment operator (++x) and the post-increment operator (x++) in C programming. Demonstrate how the results differ when both operators are used in expressions. [U][CO2]
15. Consider a scenario where you need to store the heights of players on a basketball team for analysis (for e.g., 12 players). Which data structure would you choose to store these height values in C? How would you declare and initialize this data structure? Write code to calculate and display the average height of the players on the team. [A][CO3]
16. Create a program that categorizes customer feedback ratings for a product based on the following criteria:
    - Rating '5' indicates "Outstanding."
    - Ratings '4' and '3' indicate "Satisfactory."
    - Ratings '2' and '1' indicate "Needs Improvement."

    Implement this categorization using a switch-case statement. Ensure that the program prompts the user to enter a rating and displays the corresponding feedback category. [A][CO3]

[5 x 4 = 20]

## Part C

Essay Questions
Answer **2** Questions
Each question carries **10** marks

17. Explain the problem-solving life cycle and provide an example to illustrate each step. [U][CO1]

18. Discuss one entry-controlled loop in C. Write a C program that generates and prints the Fibonacci series up to a specified number of terms using a while loop. The Fibonacci series is defined as follows:
    - The first two terms are 0 and 1.
    - Each subsequent term is the sum of the two preceding terms.

    Requirements:
    1) Prompt the user to enter the number of terms in the Fibonacci series (must be a positive integer).
    2) Calculate and print the Fibonacci series up to the specified number of terms. [A][CO3]

19. Write the purpose of using 'nested if' statements in a program. Write a nested if statement to determine the discount amount in the following scenario.

A tech store is offering discounts based on customer membership status (Member or Non-Member) and the category of electronics (Laptops, Tablets, or Accessories).
The discount rules are as follows:

| Members: | Non-Members: |
|---|---|
| Laptops: 25% discount. | Laptops: 10% discount. |
| Tablets: 15% discount. | Tablets: 5% discount. |
| Accessories: 10% discount. | Accessories: No discount. |

The program should ask the customer to input their membership status and the category of electronics, then calculate and display the discount amount. [A][CO3]

[10 x 2 = 20]